VBAT_F

GND

U5
AMS1117-3.3
VBAT_F
IN
OUT
+3V3
GND

VBAT_F
GND
J1
VBAT_IN

VBAT_F
C1
10uF
GND

+3V3
C2
10uF
GND

+3V3
C7
100nF
GND

+3V3
C8
100nF
GND

+3V3
C9
100nF
GND

+3V3
C10
100nF
VDDA

+3V3
R23
0R/100R
VDDA

VDDA
C17
1uF
GND

U2
LSM6DSV-BREAKOUT
IMU_INT1
NC
INT2
INT1
SCK
SDK
NC2
3V3
GND
SCL/SCK
SDA/SDI
CS
SAO/SDO
+3V3
GND
IMU_SCK
IMU_SDI
IMU_CS
IMU_SDO

U3
MT6701-L-01355
+3V3
GND
VDD
GND
Analog/PWM
SCL
SDA
ENC1_SCL
ENC1_SDA

U4
MT6701-R-01355
+3V3
GND
VDD
GND
Analog/PWM
SCL
SDA
ENC2_SCL
ENC2_SDA

X1
8MHz
OSC_IN
GND
VDD
OUT
GND
OSC_OUT
GND

C11
20pF
GND

OSC_OUT
C12
20pF
GND

U1
STM32F405RGT6

+3V3	19	VDD	
GND	18	GND	
VDDA	13	VDDA	
GND	12	VSSA	
+3V3	1	VBAT	
VCAP1	31	VCAP1	
VCAP2	47	VCAP2	
NRST	7	NRST	
BOOT0	60	BOOT0	
+3V3	32	VDD2	
+3V3	48	VDD3	
+3V3	64	VDD4	
GND	63	VSS2	
OSC_IN	5	PH0/OSC_IN	
OSC_OUT	6	PH1/OSC_OUT	
ENC1_SCL	14	PA0/ENC1_SCL	
ENC1_SDA	15	PA1/ENC1_SDA	
VBAT_SENSE	16	PA2/ADC1_IN2_VBAT	
VFAN_PWM	17	PA3/TIM2_CH4_VFAN	
TB_STBY	20	PA4/TB_STBY	
USER_LED	21	PA5/USER_LED	
ENC2_SCL	22	PA6/ENC2_SCL	
ENC2_SDA	23	PA7/ENC2_SDA	
PWMA	26	PB0/TIM4_CH1_PWMA	
PWMB	27	PB1/TIM4_CH2_PWMB	
IR_FL	8	PC0/ADC1_IN10_IR_FL	
IR_FR	9	PC1/ADC1_IN11_IR_FR	
IR_L	10	PC2/ADC1_IN12_IR_L	
IR_R	11	PC3/ADC1_IN13_IR_R	
IR_DL	24	PC4/ADC1_IN14_IR_DL	
IR_DR	25	PC5/ADC1_IN15_IR_DR	

PC6/TB_AIN1	37	AIN1	
PC7/TB_AIN2	38	AIN2	
PC8/TB_BIN1	39	BIN1	
PC9/TB_BIN2	40	BIN2	
PB12/SPI2_NSS_IMU_CS	33	IMU_CS	
PB13/SPI2_SCK_IMU_SCK	34	IMU_SCK	
PB14/SPI2_MISO_IMU_SDO	35	IMU_SDO	
PB15/SPI2_MOSI_IMU_SDI	36	IMU_SDI	
PB10/IMU_INT1	29	IMU_INT1	
PC10/IR_FRONT_EN	51	IR_FRONT_EN	
PC11/IR_SIDE_EN	52	IR_SIDE_EN	
PC12/IR_DIAG_EN	53	IR_DIAG_EN	
PC13/USER_BTN	2	USER_BTN	
PA11/USB_DM	44		
PA12/USB_DP	45		
PA13/SWDIO	46	SWDIO	
PA14/SWDCK	49	SWDCK	
PB3/SWO	55	SWO	
PC14	3		
PC15	4		
PB2	28		
PB11	30		
PA8	41		
PA9/USB_VBUS	42	USART1_TX	
PA10	43	USART1_RX	
PA15	50		
PD2	54		
PB4	56		
PB5	57		
PB6	58		
PB7	59		
PB8	61		
PB9	62		

+3V3
USER_BTN
SW1
USER_BTN
R6
10k
GND

U7
MP2307-MINI360-MODULE
VBAT_F
IN+
IN-
OUT+
OUT-
VFAN_3V5
GND

VCAP1
C14
2.2uF
GND

VCAP2
C15
2.2uF
GND

NRST
C15
100nF
GND

BOOT0
R2
10k
GND

IR_FRONT_EN
Q8
AO3400_FRONT
R19
10k
GND

+3V3
D2
SFH4545_FL

+3V3
D3
SFH4545_FR

R7
33

R8
33

IR_SIDE_EN
Q9
AO3400_SIDE
R20
10k
GND

+3V3
D4
SFH4545_L

+3V3
D5
SFH4545_R

R9
33

R10
33

IR_DIAG_EN
Q10
AO3400_DIAG
R21
10k
GND

+3V3
D6
SFH4545_DL

+3V3
D7
SFH4545_DR

R11
33

R12
33

+3V3
Q2
TEFT4300_FL
IR_FL
R13
10k
GND

+3V3
Q3
TEFT4300_FR
IR_FR
R14
10k
GND

+3V3
Q4
TEFT4300_L
IR_L
R15
10k
GND

+3V3
Q5
TEFT4300_R
IR_R
R16
10k
GND

+3V3
Q6
TEFT4300_DL
IR_DL
R17
10k
GND

+3V3
Q7
TEFT4300_DR
IR_DR
R18
10k
GND

R3
22k
VBAT_F
VBAT_SENSE

R4
10k
GND

+3V3
NRST
R1
10k

VAC_MOTOR-
VFAN_PWM
Q1
AO3400_VAC
R22
10k
GND
GND

R5
330
GND

D1
LED_BLUE
USER_LED

MOT_AO1
MOT_AO2
J4
MOT_LEFT

C13
100uF
VBAT_F
GND

MOT_BO1
MOT_BO2
J5
MOT_RIGHT

U8
TB6612FNG-BREAKOUT
VBAT_F
+3V3
GND
MOT_AO1
MOT_AO2
MOT_BO2
MOT_BO1
GND
VM
VCC
GND
AO1
AO2
BO2
BO1
GND2
PWMA
AIN2
AIN1
STBY
BIN1
BIN2
PWMB
GND
PWMA
AIN2
AIN1
TB_STBY
BIN1
BIN2
PWMB
GND

VFAN_3V5
VAC_MOTOR-
J3
VAC_MOTOR

VAC_MOTOR-
D8
SS14_FLYBACK
VFAN_3V5

+3V3
SWDCK
SWDIO
SWO
USART1_TX
USART1_RX
GND
J7
SWD_UART_SWO

Drive: 2x Coreless 1718 @ 2S LiPo | Vacuum: Coreless 1020 @ 3.5V Mini360 Buck Module
MCU: STM32F405RGT6 | IMU: LSM6DSV Green Breakout | ENC: 2x MT6701 01355 Breakout | DRV: TB6

ZAIR / Sri Lanka IIT Micro Maze 3.0

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Title: 4WD Vacuum Micromouse â€“ ZAIR

Size: A3	Date: 2026-08-16	**Rev: 11.0**
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